

Development and Experimental Applications of a Direct Digital Synthesis (DDS)-Based Signal Generation Platform

(A Low-Cost RF Signal Generator and Antenna Analyzer using Arduino Uno and AD9850)

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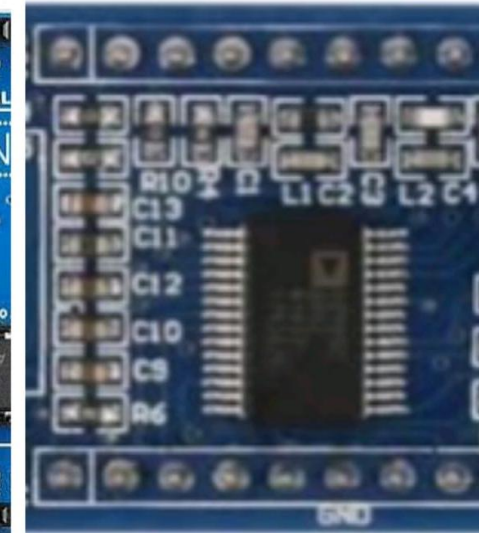
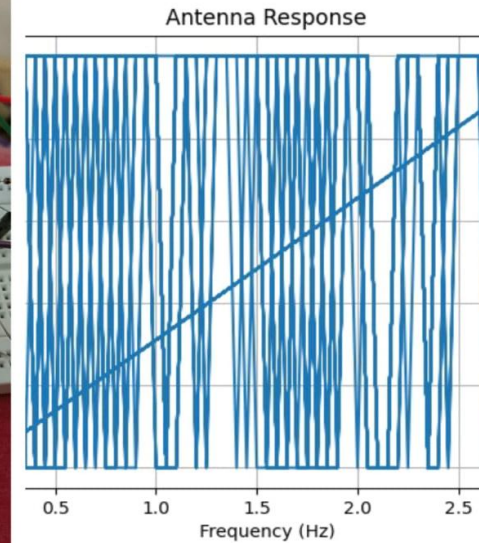
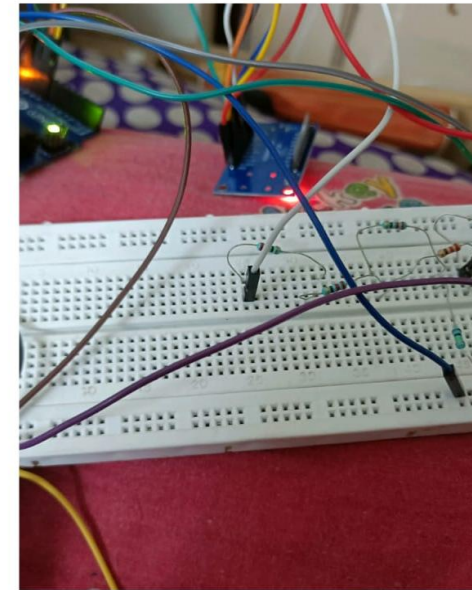
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Motivation, Problem Statement and Objectives

MOTIVATION

Modern RF laboratories require instruments like

- RF Signal Generator
- Spectrum Analyzer
- Baseband modulators
- Antenna Analyzer

These instruments typically cost **₹30,000 to several lakhs**, making them inaccessible to undergraduate laboratories.



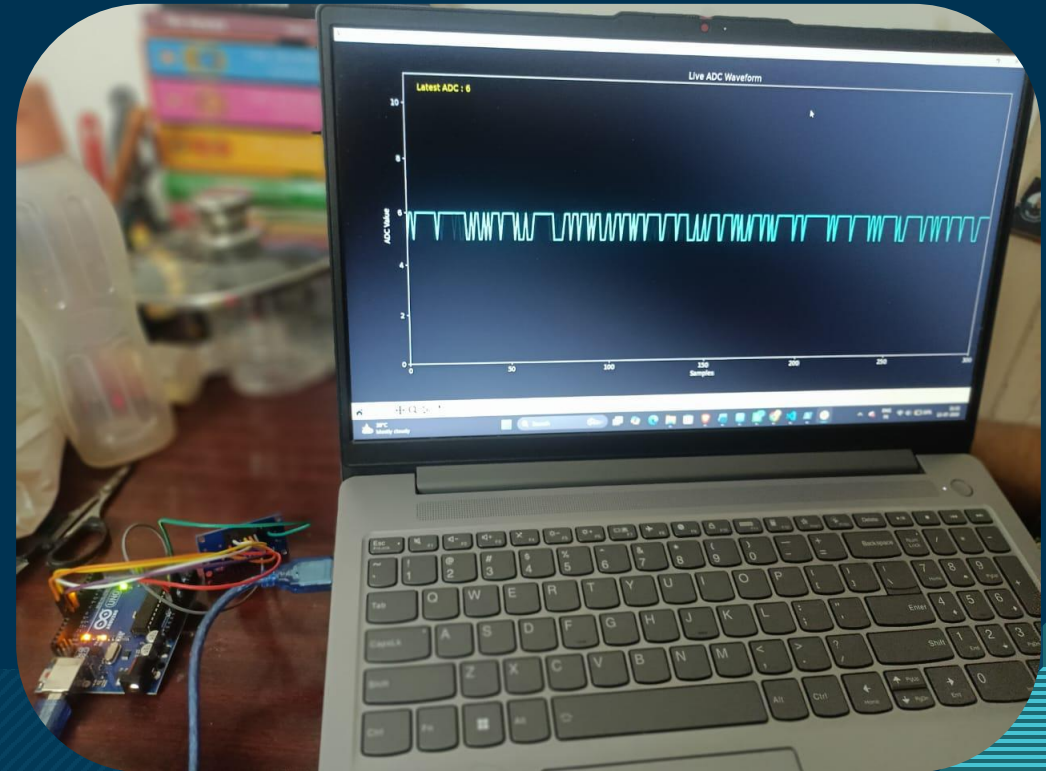
PROBLEM STATEMENT

Develop a **low-cost programmable RF platform** capable of

- RF signal generation
- Antenna analysis
- SWR measurement
- Future digital communication experiments
- Impedance analysis

OBJECTIVES

- Develop DDS based RF source
- Interface AD9850 with Arduino
- Perform frequency sweep
- Measure reflected RF signal
- Calculate Reflection Coefficient and SWR
- Build a scalable RF experimentation platform
- Develop Python visualization software



Proposed DDS-Based RF Experimental Platform

The generated RF signal is utilized in **three major applications**:

Application I – DDS-Based Digital Communication (Proposed)

- Programmable RF carrier generation
- Phase-controlled modulation
- Future implementation of BPSK and QPSK transmitters
- Foundation for Software Defined Radio (SDR) applications

Arduino Uno

Frequency Tuning Word(FTW)

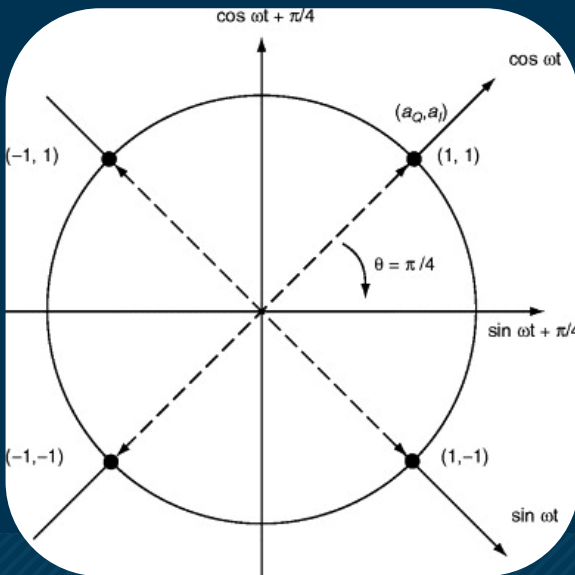
AD9850 DDS Module

Programmable RF Signal Source

DDS Digital Communication (BPSK/QPSK)

Wire Antenna Analyzer (SWR & Γ)

RLC Impedance Analyzer (R, L, C, Z)

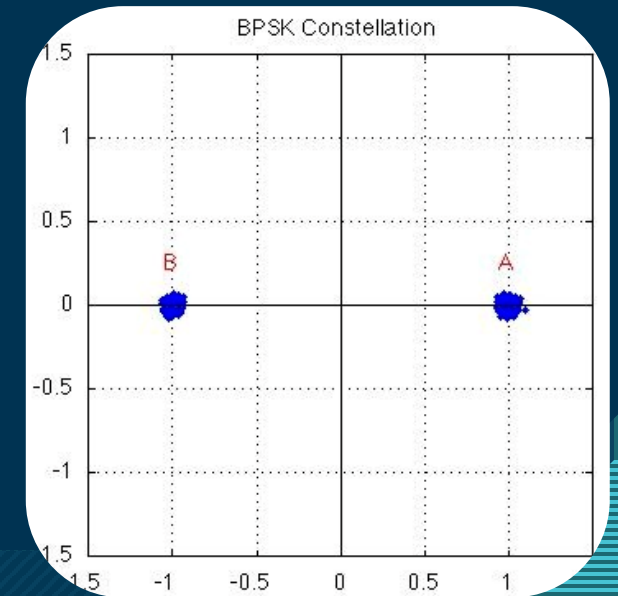


Application II – Wire Antenna Analyzer (Implemented)

- Frequency sweep of the Antenna Under Test (AUT)
- Reflection coefficient measurement
- Standing Wave Ratio (SWR) calculation
- Python-based graphical visualization of antenna characteristics

Application III – RLC Impedance Analyzer (Implemented/Under Development)

- Frequency-dependent impedance measurement
- Resistance (R), Reactance (X), and Impedance (Z) calculation
- Analysis of resonant frequency and component characteristics
- Graphical impedance visualization using Python



Direct Digital Synthesis (DDS)

DDS generates waveforms digitally rather than using analog oscillators.

Main blocks

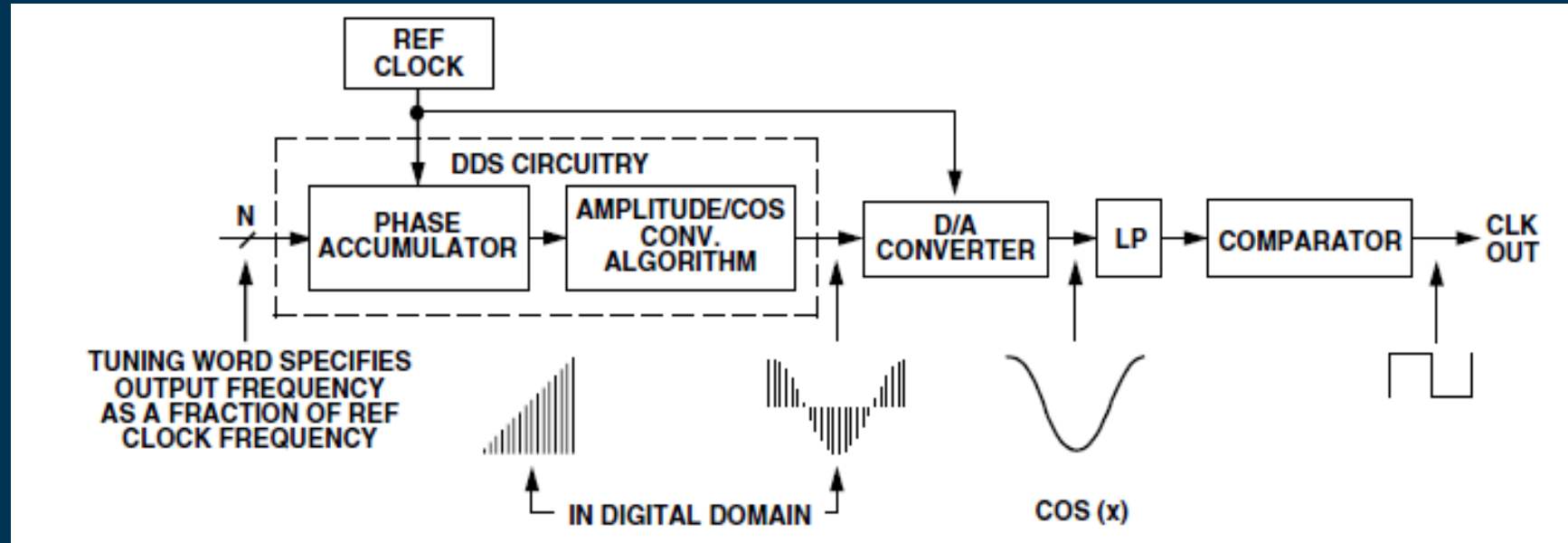
- Reference Clock
- Phase Accumulator
- Phase-to-Amplitude Converter
- DAC
- LPF

Frequency Equation $f_{out} = \frac{FTW \times f_{clk}}{2^{32}}$

- Arduino computes FTW
- FTW loaded into AD9850
- AD9850 generates sine wave

Advantages

- 32-bit tuning
- 0.029 Hz resolution
- Fast switching
- Excellent stability
- Phase continuity



AD9850 DDS Module

Internal blocks

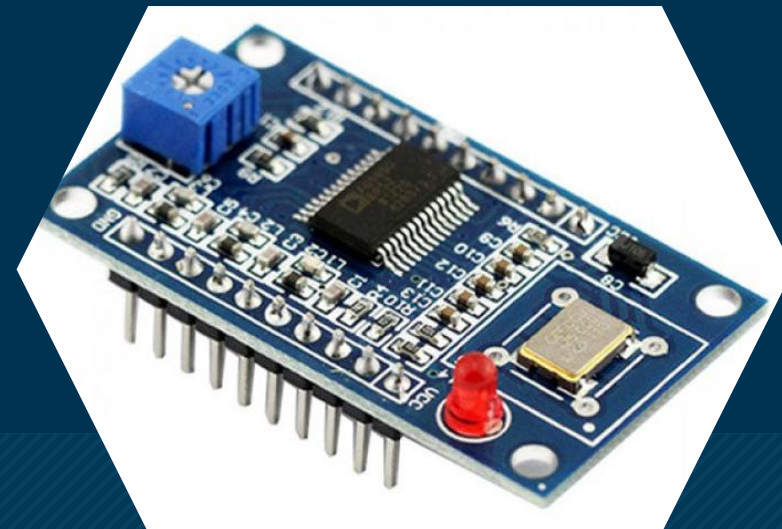
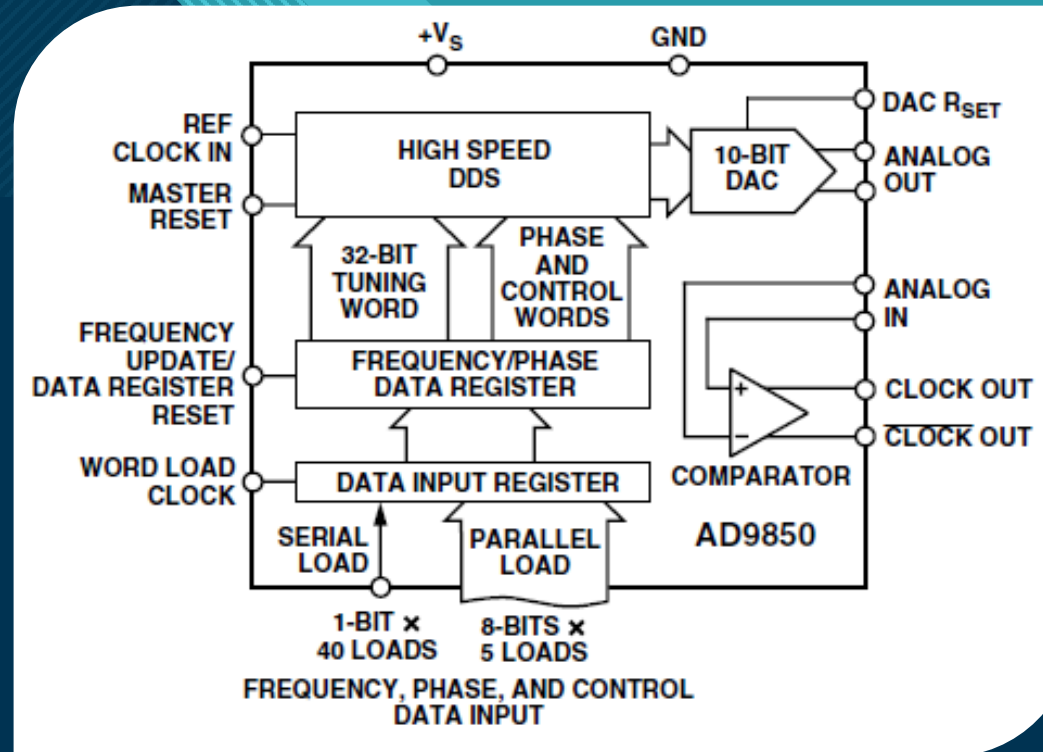
- 32-bit Phase Accumulator
- ROM Lookup Table
- 10-bit DAC
- Comparator
- Serial Interface

Specifications

Parameter	Value
Clock	125 MHz
DAC	10 bit
Resolution	0.029 Hz
Output	≈62.5 MHz
Programming	Serial

Why use AD9850

- Cheap
- Accurate
- Arduino compatible
- Widely used in SDR



Implemented Application I

RF Signal Generator

Arduino sends
Frequency Tuning Word

DDS synthesizes
waveform

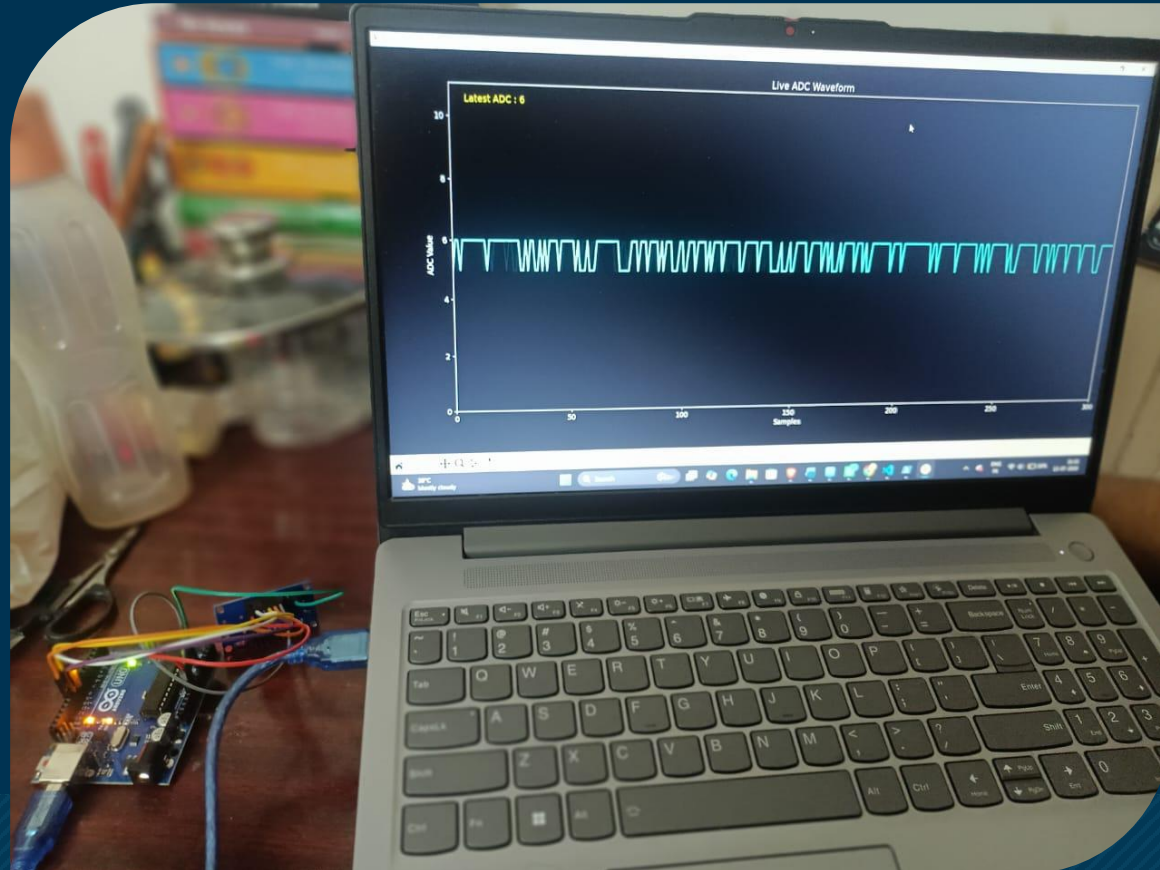
DAC converts digital
samples

LPF removes
harmonics

Stable RF output

Possible applications

- Filter testing
- Clock generation
- RF experiments
- Communication laboratory



Digital Phase Shift Keying using DDS

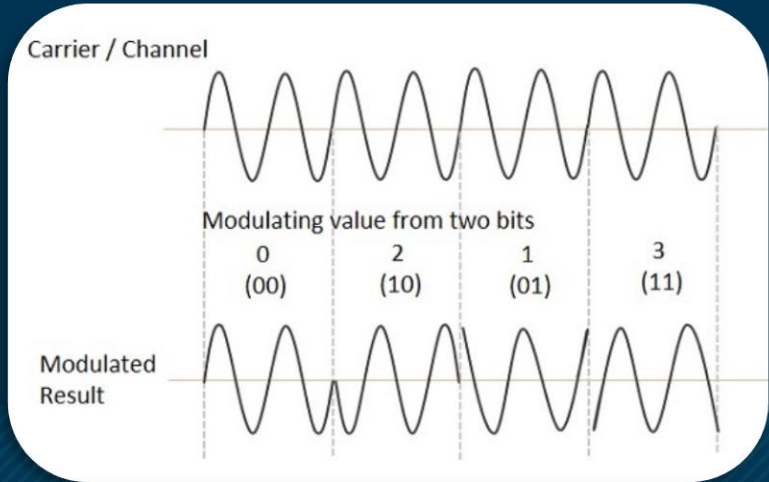
Phase Shift Keying (PSK)

Digital modulation where information is conveyed by changing only the **carrier phase**, while amplitude and frequency remain constant.

Binary Phase Shift Keying (BPSK)

- Simplest form of digital phase modulation.
- Uses **two phase states** separated by 180°.
- Each transmitted symbol carries **one bit**.

Binary Data	Carrier Phase
1	0°
0	180°



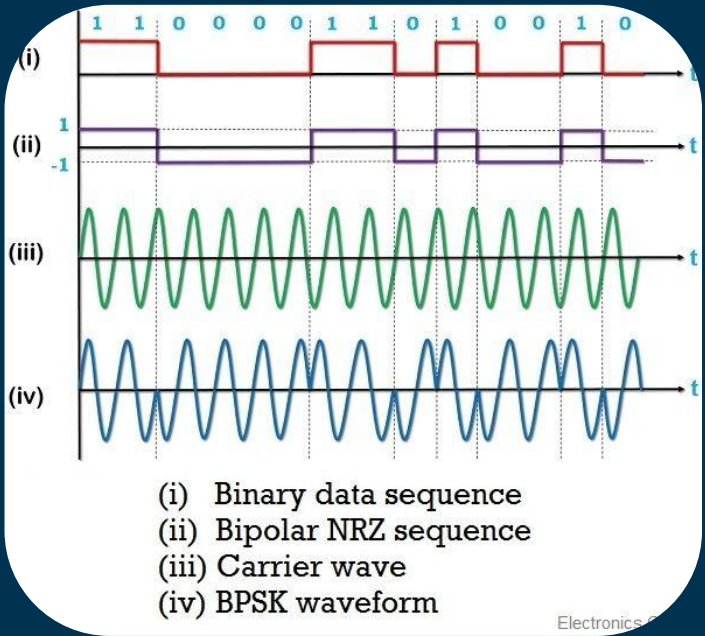
Advantages

- High noise immunity
- Simple transmitter and receiver
- Widely used in GPS, satellite communication, and telemetry

Quadrature Phase Shift Keying (QPSK)

- Extension of BPSK.
- Uses **four equally spaced carrier phases**.
- Each symbol represents **two bits**, doubling the data rate without increasing bandwidth.

Bits	Phase
00	45°
01	135°
11	225°
10	315°



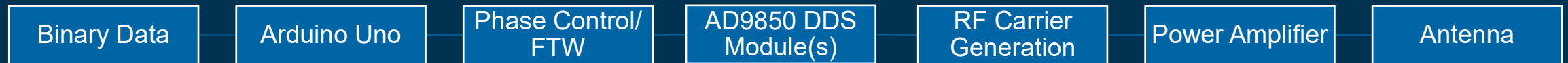
Advantages

- Higher spectral efficiency
- Better bandwidth utilization
- Commonly used in LTE, Wi-Fi, DVB, and satellite communication

Proposed DDS-Based Transmitter

The developed DDS platform can be extended into a programmable digital transmitter.

- **Arduino Uno** generates the input bit stream.
- The **AD9850 DDS module** generates the RF carrier.
- The Arduino updates the **Frequency Tuning Word (FTW)** and **phase information** to produce the required phase shifts.
- For **BPSK**, a single AD9850 can generate carriers with **0° and 180° phase shifts**.
- For **QPSK**, two synchronized AD9850 modules can generate **In-phase (I)** and **Quadrature (Q)** carriers separated by **90°**, which are combined to produce the QPSK waveform.
- This approach provides **software-controlled modulation**, rapid phase switching, and excellent frequency stability.



Wire Antenna Analyzer and SWR Measurement

Theory of Antenna Analysis

Efficient power transfer between a transmitter and an antenna requires **impedance matching**. When the antenna impedance differs from the characteristic impedance of the transmission line (typically **50 Ω**), a portion of the incident RF signal is reflected back toward the source. These reflections reduce the transmitted power and degrade communication system performance.

To evaluate antenna performance, three important parameters are measured:

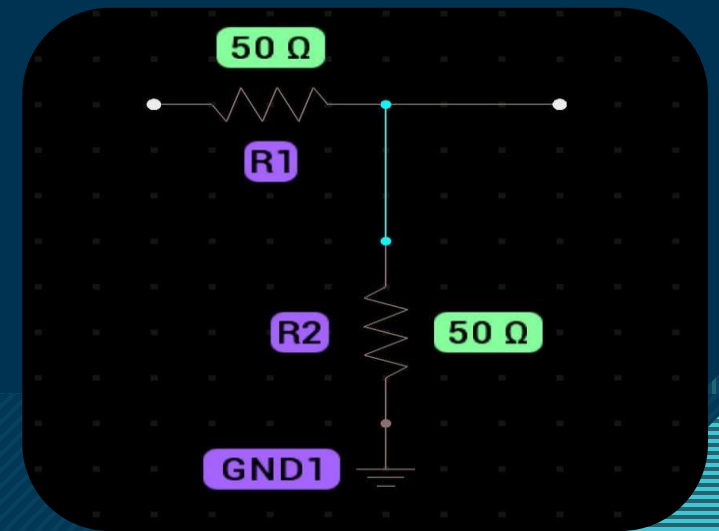
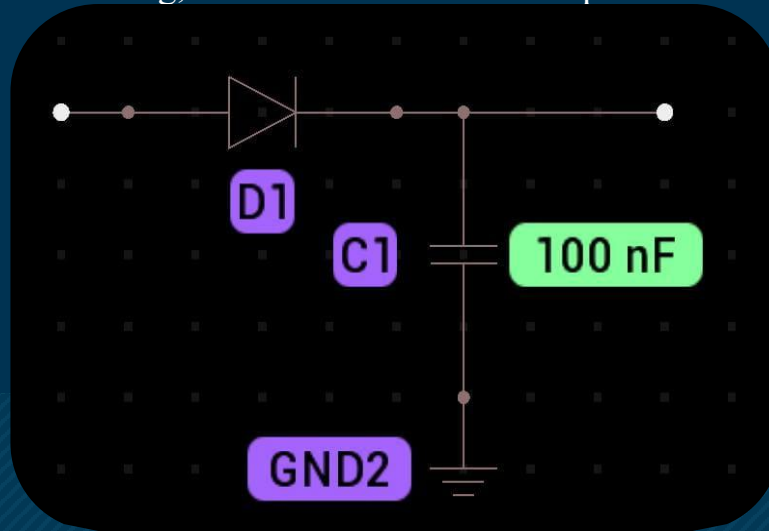
Reflection Coefficient (Γ)

The Reflection Coefficient represents the ratio of the reflected signal to the incident signal and indicates the degree of impedance mismatch.

$$\Gamma = \frac{V_r}{V_i}$$

where,
 V_r = Reflected Voltage
 V_i = Incident Voltage

A value of $\Gamma = 0$ indicates perfect impedance matching, while $\Gamma = 1$ indicates complete reflection.



Wire Antenna Analyzer and SWR Measurement

Standing Wave Ratio (SWR)

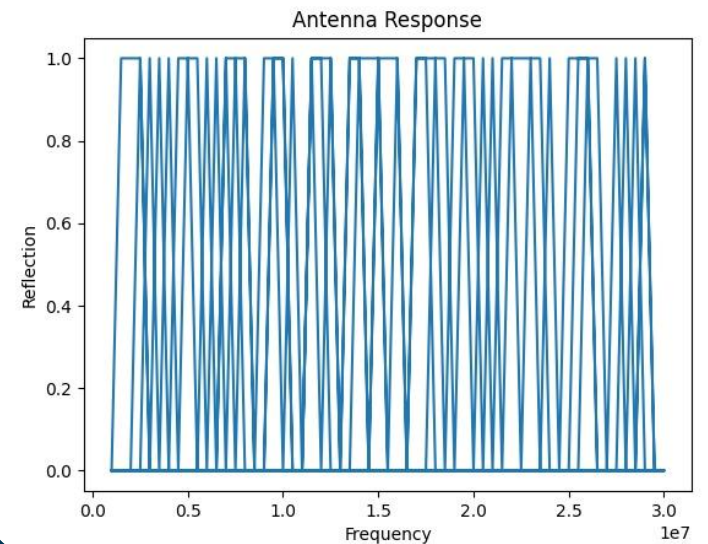
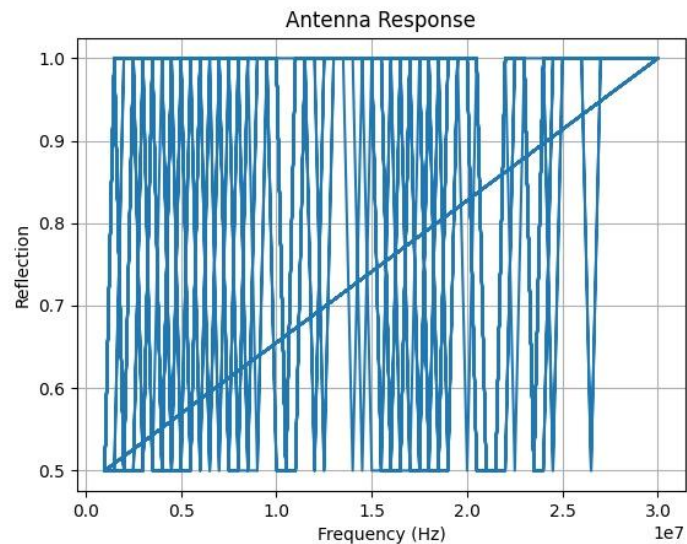
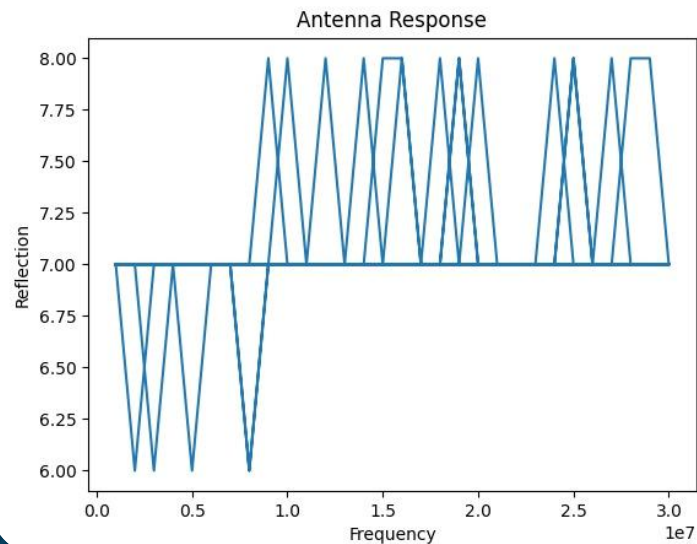
Standing Wave Ratio (SWR) is a measure of the efficiency of power transfer between the transmission line and the antenna.

$$SWR = \frac{1 + |\Gamma|}{1 - |\Gamma|}$$

Interpretation:

- **SWR = 1 : 1** → Perfect antenna matching
- **SWR < 2 : 1** → Good antenna performance
- **High SWR** → Significant power reflection and mismatch

Monitoring SWR allows the resonant frequency and impedance characteristics of the antenna to be determined.



DDS-Based Wire Antenna Analyzer

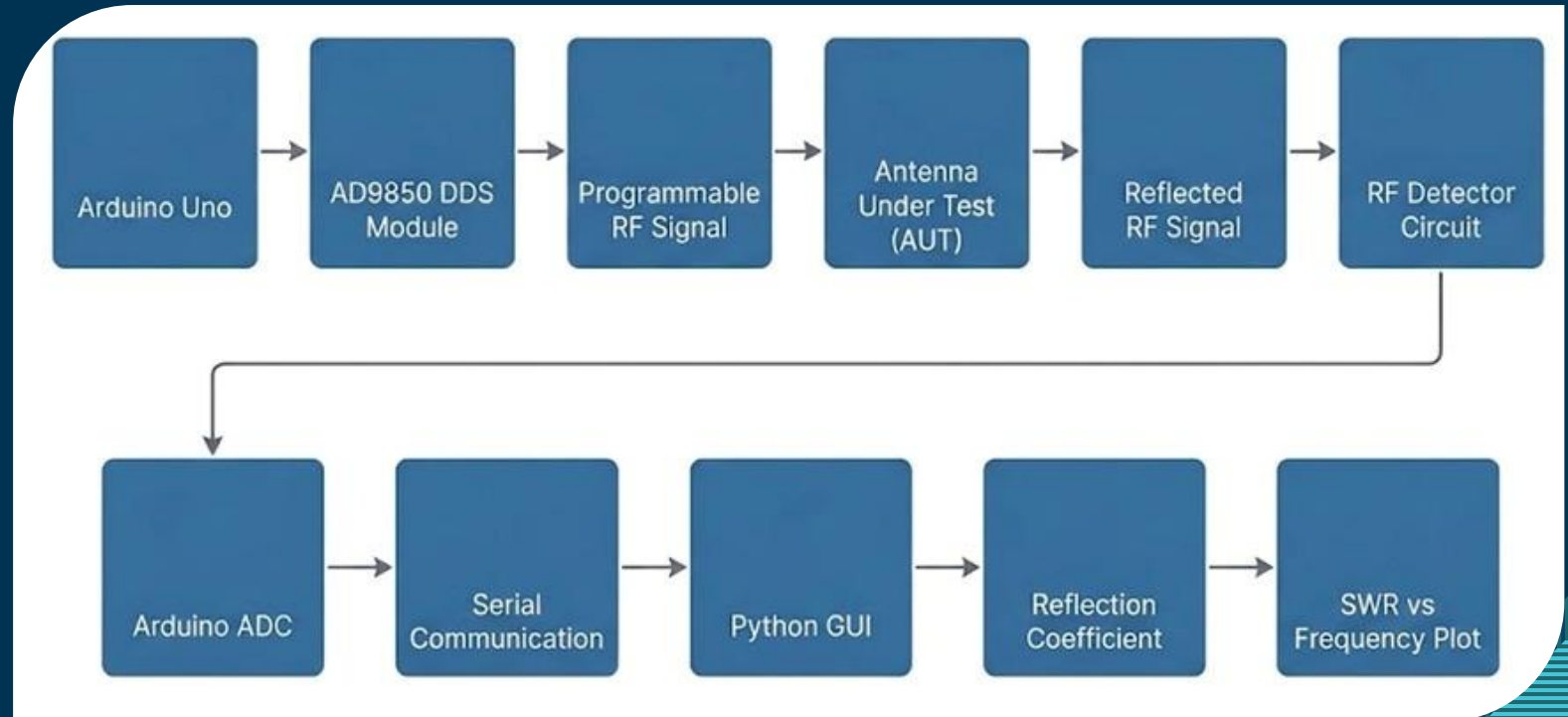
The developed system utilizes the **Arduino Uno** and **AD9850 DDS module** to generate a programmable RF signal over a selected frequency range.

An RF detector measures the reflected signal caused by impedance mismatch and converts it into a proportional DC voltage. The Arduino acquires this voltage using its **10-bit ADC** and transmits the data to a computer through serial communication.

A Python-based application then:

- Calculates the Reflection Coefficient (Γ)
- Computes the Standing Wave Ratio (SWR)
- Performs frequency sweep analysis
- Generates real-time SWR vs Frequency plots using Matplotlib

This implementation provides a **low-cost alternative to commercial antenna analyzers**



DDS-Based RLC Impedance Analyzer

Theory of Impedance Measurement

The electrical behavior of an RLC component varies with frequency. By exciting the component with a **programmable RF signal** and measuring its response, the circuit's impedance can be determined.

The total impedance is given by

$$Z = R + jX$$

where,

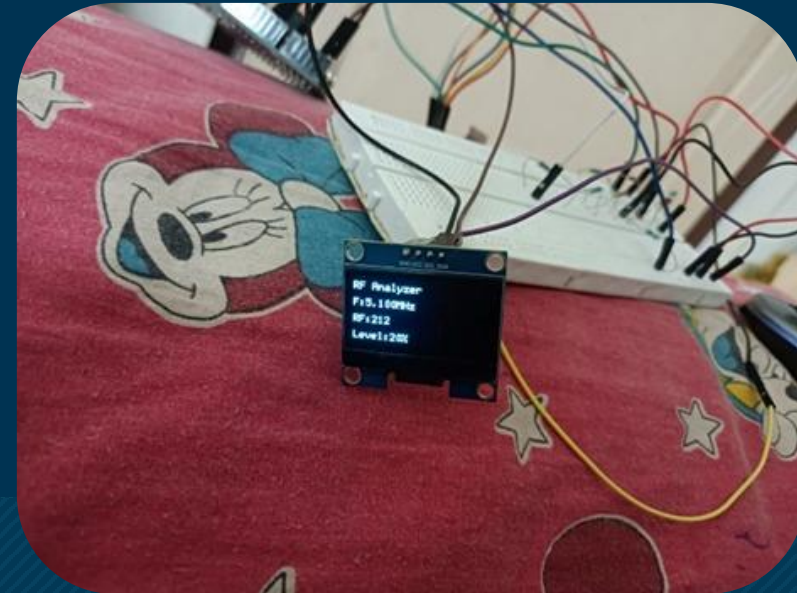
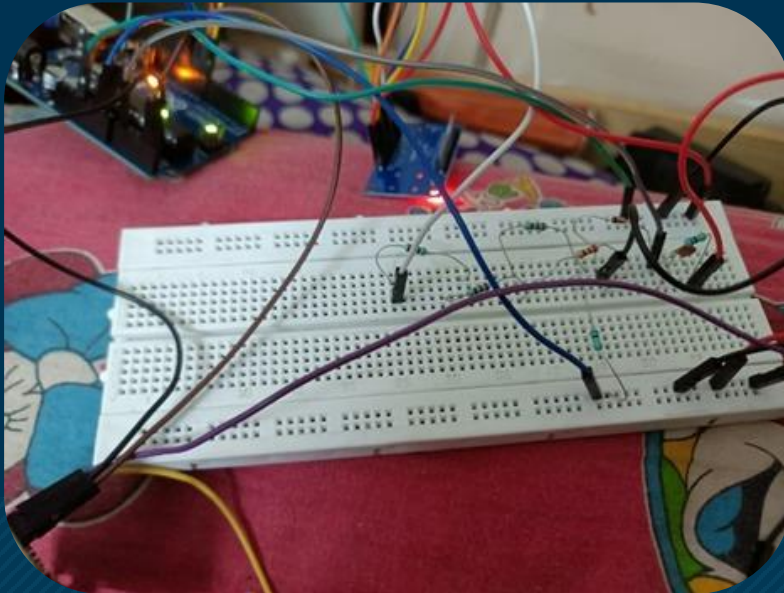
R = Resistance

X = Reactance (Inductive or Capacitive)

For a series RLC circuit,

$$X = X_L - X_C = \omega L - \frac{1}{\omega C}$$

By sweeping the input frequency, the resonant frequency and frequency response of the component can be analyzed.



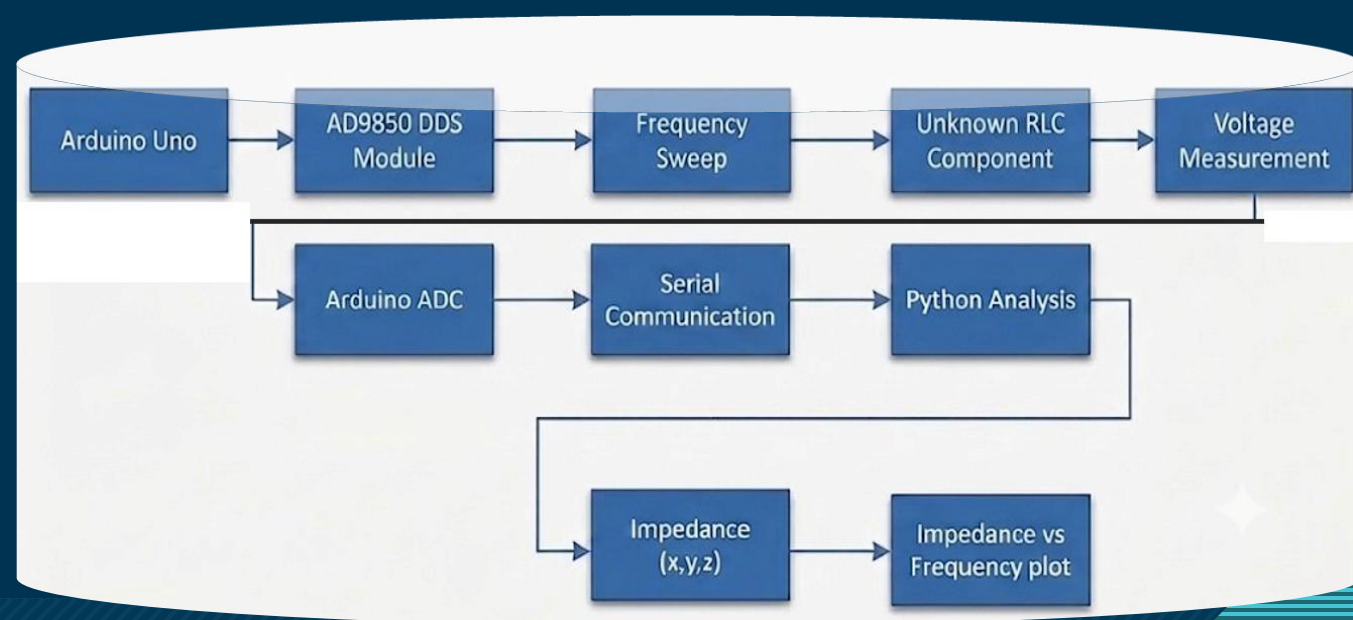
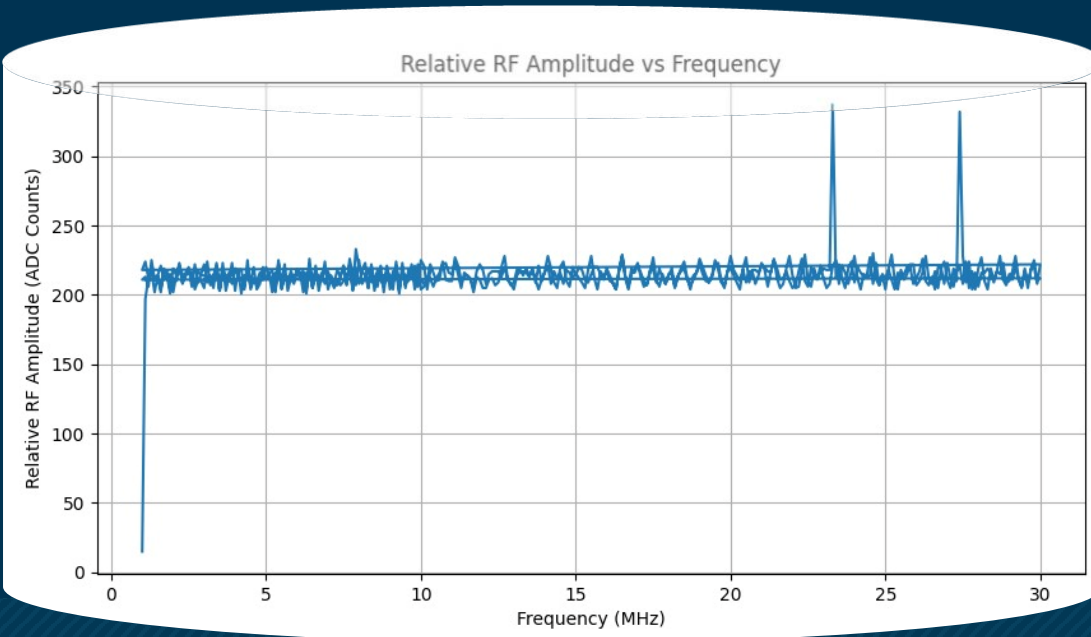
DDS-Based Implementation

The proposed analyzer uses the **Arduino Uno** and **AD9850 DDS module** to generate a programmable frequency sweep across the RLC component.

The measured voltage is acquired by the Arduino and processed using a Python application to calculate:

- Resistance (R)
- Reactance (X)
- Impedance (Z)
- Resonant Frequency

The results are displayed graphically as **Impedance vs Frequency**, enabling characterization of electronic components using the same DDS hardware platform.



Performance Analysis

Comparison Table:

Parameter	Commercial RF instruments	Proposed DDS-based Platform
Approximate Cost	₹50,000 – ₹3,00,000+	< ₹3,000
Signal Source	Analog Oscillator / PLL / DDS	AD9850 Direct Digital Synthesis (DDS)
Frequency Sweep	Built-in Hardware Sweep	Software-Controlled Sweep
Frequency Resolution	Instrument Dependent	32-bit DDS Resolution (~0.029 Hz)
Frequency Programming	Front Panel / Menu Based	Arduino-Controlled Digital Programming
Phase Control	Manual / Limited Digital Control	Real-Time Software Phase Control
Waveform Configuration	Fixed Instrument Functions	Software Configurable
Automation Capability	Limited External Control	Full Arduino & Python Integration
Expandability	Hardware Dependent	Modular and Highly Expandable
Digital Communication Support	Optional / High-Cost Models	Suitable for BPSK, QPSK & SDR Development
Software Support	Proprietary Software	Open-Source Python Platform

Performance Analysis of the Proposed DDS-Based RF Platform

Technical Advantages of DDS-Based Signal Generation

- **32-bit AD9850 DDS** provides precise digital RF signal generation.
- **Arduino Uno** enables software-controlled frequency and phase tuning.
- Programmable frequency sweep for flexible RF testing.
- High frequency resolution of **~0.029 Hz**.
- Python-based automation for data acquisition and visualization.
- Modular and low-cost architecture suitable for RF experiments

Current Limitations:

- Frequency accuracy depends on the stability of the **125 MHz reference oscillator**.
- RF detector requires calibration for accurate SWR measurements.
- Output power is limited and may require an external RF amplifier.
- Designed primarily for educational and laboratory applications rather than precision RF instrumentation.

Advantages for Digital Communication

The proposed platform can be extended to support:

- Binary Phase Shift Keying (**BPSK**)
- Quadrature Phase Shift Keying (**QPSK**)
- Frequency Shift Keying (**FSK**)
- Software Defined Radio (**SDR**)
- Frequency Hopping Communication
- Adaptive Digital Communication Experiments

Future Scope

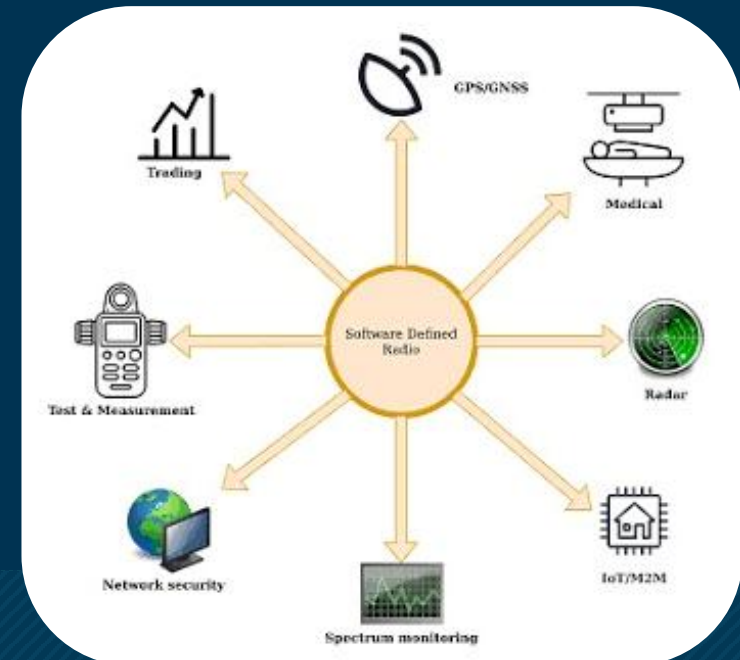
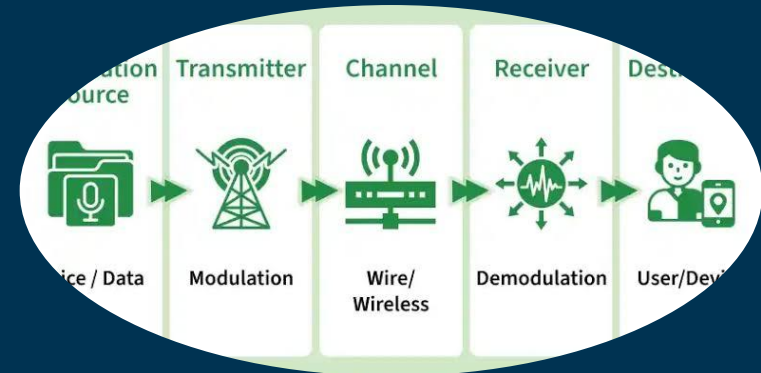
Advanced Communication
BPSK, QPSK, FSK
Higher-order Mod.
Phase Control

Enhanced RF Measurement
Return loss
Smith Chart
VNA integration

Portable Instrument
TFT Display
Battery Powered
Wireless Connectivity

Intelligent Automation
Auto Sweep
AI Analysis
Cloud Logging

SDR Integration
RTL-SDR
HackRF
Spectrum Analysis
Research



Conclusion

- ✓ Developed a low-cost DDS-based RF signal generation platform using the AD9850 DDS module and Arduino Uno.
- ✓ Successfully implemented wire antenna analysis using Reflection Coefficient (Γ) and Standing Wave Ratio (SWR) measurements.
- ✓ Designed a Python-based GUI for real-time frequency sweep, data acquisition, and visualization.
- ✓ Demonstrated the flexibility of the DDS platform for digital communication experiments such as BPSK, QPSK, and future SDR applications.
- ✓ Achieved a software-controlled, modular, and scalable architecture suitable for educational, laboratory, and research applications.

References

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